MILITARY SPECIFICATION

CHARTS, OPERATING INSTRUCTIONS,

FOR SHIP OR SHORE ELECTRONIC EQUIPMENT

1. SCOPE

1.1 This specification covers the preparation of operating instruction charts to be mounted on or adjacent to installed electronic equipment to enable an operator inexperienced with the particular equipment to start and operate the equipment.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

L-V-351 - Vinyl Chloride Polymer and Copolymer Rigid Sheets.

MILITARY

MIL-P-78 - Plastic-Material, Laminated, Thermosetting (for Designation Plates).

NAVY DEPARTMENT

General Specifications for Inspection of Material.
16B16(RE) - Books, Instruction, for Ship or Shore Electronic Equipment.
16E4 - Electronic Equipment, Naval Ship and Shore: General Specification.

STANDARDS

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings.
MIL-STD-15 - Electrical and Electronic Symbols.
MIL-STD-16 - Reference Designations for Electrical and Electronic Symbols.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Definitions.-

3.1.1 Control station.- For the purposes of this specification, a control station is defined as a station (operating position) located on ship or shore at which an operator can place a system, set, group, or major unit into operating condition. For example, switchboard location, PPI (plan position indicator) repeater station, antenna remote control station.

3.1.2 In addition, definitions of Specification 16E4 shall apply.

3.2 Materials.- The materials shall be as specified herein. The best material commercially available for the purpose shall be used when a definite material is not designated.

3.3 General.- The instruction chart shall outline the operating procedures and shall be written entirely for the operator in accordance with Specification 16B16(RE) (see fig. 1). It shall include a step-by-step description to enable the operator inexperienced with the particular equipment to start and operate the equipment without reference to instruction books. The names of all controls which are titled or labeled on the equipment and referenced shall be printed exactly as they appear thereon. Only those controls normally required for use during operation shall be referenced. Instructions shall be complete and in logical sequence. Where provisions exist for local or remote operation of the equipment, operating instructions shall be complete for each type of control station.

3.4 Approval.- Four copies of the charts in manuscript form shall be submitted to the bureau or agency concerned for approval of content and assignment of a publication number. In addition, one copy, shall be submitted to the Government inspector. These manuscripts should be submitted as early as possible to insure shipment of charts with the first production equipment.

3.5 Security classification.- The instruction charts shall not include material classified higher than RESTRICTED. Essential information of a security classification higher than restricted shall be brought to the attention of the bureau or agency concerned.

3.6 Mounting.- Instruction charts shall be provided with mounting holes reinforced, if necessary, by eyelets (see fig. 1). Usually the charts will be installed in prominent view of an operator in normal operating position. Where space on the equipment is not available, charts shall be temporarily attached to the equipment for later installation by installing activities.

3.7 Number of charts.- A separate chart shall be provided for each control station. If possible, complete operating instructions shall be included on a single chart for each control station. If this is impossible, two or more successive charts may be employed. These shall be bound together by a suitable hinge such as plastic spiral or ring binding.

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3.7.1 Two sets of charts shall be furnished for each equipment of which one set shall be supplied with the equipment. In addition, a bulk quantity of 200 sets of charts shall be provided, unless otherwise specified in the contract or order, of which 10 sets shall be furnished to the bureau or agency concerned. At the same time, shipping instructions for the balance shall be requested from the bureau or agency concerned.

3.8 Size.- Charts shall be of the smallest practicable size and shall in no case exceed 9 by 12 inches.

3.9 Layout.- Futura ultra bold type, of suitable size and contrast shall be used in titles; Garamond bold type shall be used in the text. Alternate type faces as specified in Specification 16B16(RE) may be used. Where space is available, a minimum of 10 point type shall be used. A border of 1/4 inch or more shall be provided on each edge of the instruction chart. Two columns of text may be used if desired. Instructions shall appear on one side of the chart only, unless specifically approved by the bureau or agency concerned.

3.10 Legibility.- Charts shall retain legibility during the expected service life of the equipment.

3.11 Illustrations.- Illustrative material, either photographic or line drawings, shall be used if required for clarity. Controls and reference designations referenced in the instructions, but not readily identified by casual inspection of the equipment, shall be clearly identified.

3.12 Materials and processes.- One of the following materials and processes shall be used for instruction charts, unless authorized by the bureau or agency concerned:

(a) Lithographed and laminated on white plastic sheeting conforming to type GCP-H of Specification MIL-P-78.
(b) Printed on a good grade white book paper, and laminated between two sheets of clear transparent vinyl resin, conforming to class 1 of Specification L-V-351, which are bonded together to seal against moisture.
(c) Lithographed or printed on aluminum, nickel-copper alloy, or corrosion-resistant steel plate. (This method is to be used only if specifically approved by the bureau or agency concerned.)
(d) Photo-etched on aluminum.

3.13 Master reproducibles.- Master reproducible drawings and negatives of operating instruction charts shall be furnished to the bureau or agency concerned. Shipping instructions will be supplied upon request.

3.14 Contents.- Instruction charts shall contain the following: (see fig. 1).

3.14.1 Title.- The title shall identify the equipment and major unit to which the instruction chart applies and the control station intended. It shall be centered at the top of the chart. In addition, the chart shall have a publication number, assigned by the bureau or agency concerned, located in the upper left hand corner. The security

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classification, if applicable, followed by the words "Security Information", shall be placed in the upper right hand corner. The security classification shall appear at the bottom center. If two or more charts are required at any one control station, the charts shall be numbered consecutively in the lower right hand corner such as "Chart 1 of_____", "Chart 2", "Chart 3".

3.14.2 Caution notices.- Caution notices, if required, shall be provided in a prominent manner.

3.14.3 Location of important controls and switches.- The location of important auxiliary switches and controls (such as main power and antenna switches), not mounted on the equipment itself shall be specified. If the location cannot be determined in a general way, space shall be reserved on the chart for the using activity to note such locations in ink or indelible pencil.

3.14.4 Text.- The text shall consist of a straightforward, step-by-step procedure for starting and operating the equipment from the control station in question. The procedure should be understandable to an operator who is seeing the equipment for the first time. The text shall conform to Specification 16B16(RE). All controls shall be referred to exactly as they are marked on the equipment (such as spelling, abbreviations, capitalization). Reference to items of the equipment shall be by type designation, standard nomenclature, or reference designation, as applicable. Abbreviations, graphic symbols, and reference designations for electronic parts shall be in accordance with Standards MIL-STD-12, MIL-STD-15 and MIL-STD-16, respectively.

3.15 Workmanship.- Workmanship shall be first class in every respect.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The Government inspector shall determine that the charts conform to all the requirements specified herein and that the charts are as approved by the bureau or agency concerned.

4.2 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in Specification 16B16(RE).

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any

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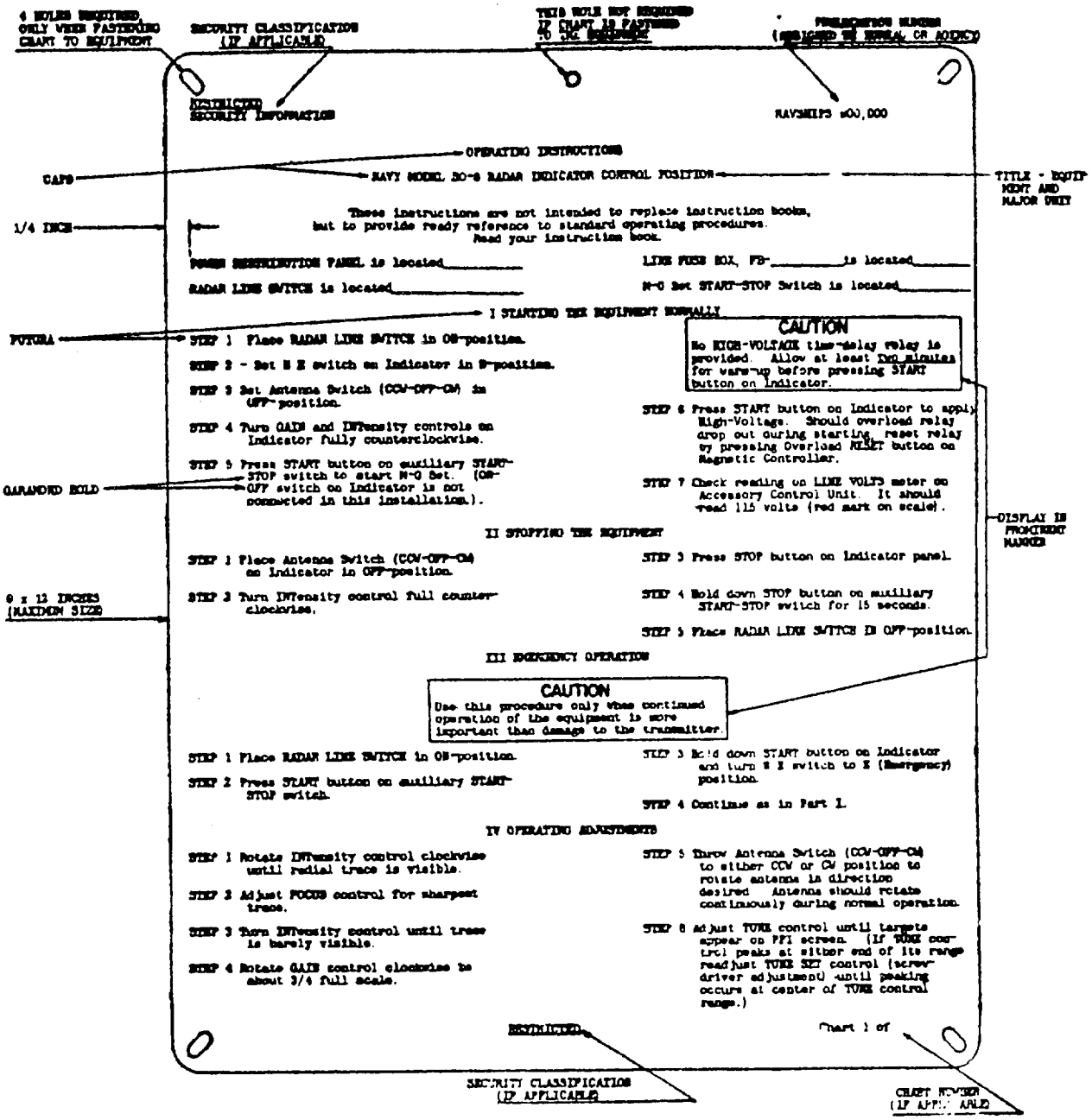


FIGURE 1 - SAMPLE CHART

(THIS FIGURE IS MARKED RESTRICTED FOR DEMONSTRATION PUSPOSES ONLY.)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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